

Wearable computing and the remembrance agent

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This paper gives an overview of the field of wearable computing. It covers the key differences between wearables and other portable computers, and discusses issues with the design and application for wearables. There then follows a specific example, the wearable remembrance agent — a proactive memory aid. The paper concludes with discussion of future directions for research and applications inspired by using the prototype.

1. Introduction

‘I am a new field engineer issued with the latest support equipment — a wearable computer that I clip to my belt, with a throat microphone, an earpiece and a head-up display that looks like a pair of normal glasses. My first job is to do a repair at a large building complex which is unfamiliar to me. As I approach the complex my wearable whispers in my ear, reminding me I have to report to a certain building first and gives appropriate directions from my current position. As I check in, my head-up-display reminds me about a recent fault that was fixed by another engineer, so I enquire if the fix has been suitable. As I get to the equipment for repair, my wearable notifies me of a virtual reminder that was left there by the installation engineer. I read the reminder on my display before starting work. As I start to work, notes about similar faults are brought to my attention which makes it easier to track down the problem. In this case I know I will need to liaise with one of my colleagues at some point who is a specialist in this area. There are a few to whom I could talk and from the background ‘noise’ from the wearable I know who is available and when. I have finished the repair work, and it is interesting that the wearable is showing me that there were a number of faults that happened at about the same time of day indicating some other, common cause which needs to be tracked.’

This scenario may sound some way off in practical terms, but it is not. All the technologies needed to support it are available. It is the aim of this paper to describe some of these technologies in more detail and give an indication of their current status. The paper will start by describing features available in wearable computers that are not available in current laptops or personal digital assistants (PDAs). It will then go on to describe a number of other general application areas for wearables and current wearable technologies and design needs. This is followed

by a description of the remembrance agent (RA), a wearable memory aid that continually reminds the wearer of potentially relevant information based on the wearer’s current physical and virtual context. Finally, the paper discusses extensions that are being added to the current prototype system.

2. What are wearable computers?

A wearable computer is not simply a computer that the user wears, it is a host for an application or set of applications. It is obviously appropriate that if one can have powerful local computing power and advanced sensors then there are many potential applications that can be developed. This was not the case a few years ago, the computing power required could only be achieved with a workstation-sized machine and therefore wearables were kept in the realms of science fiction. Now they are practical propositions.

The fuzzy definition of a wearable computer is that it is a computer that is always with the user, is comfortable and easy to keep and use, and is as unobtrusive as clothing. However, this ‘smart clothing’ definition is unsatisfactory when the details are considered. Most importantly, it does not convey how a wearable computer is any different from a very small palm-top. A more specific definition is that wearable computers have many of the following characteristics.

- Are portable while operational

The most distinguishing feature of a wearable is that it can be used while walking or otherwise moving around. This distinguishes wearables from both desktop and laptop computers, but does not distinguish them from portable phones.

- Utilise sensors

In addition to normal user inputs, a wearable should have sensors for the physical environment such as location (for example GPS), cameras, microphones, temperature, humidity, movement, etc.

- Enable hands-free use

Military and industrial applications for wearables especially emphasise their hands-free aspect, and concentrate on speech input and heads-up display or voice output. Other wearables might also use chording-keyboards, dials, and joysticks to minimise the tying up of a user's hands. Other applications use context-based information provided by sensors that do not rely on any direct user input, but are guided by the environment.

- Can be proactive

A wearable should be able to convey information to the user even when not in active use. For example, if your computer wants to let you know you have new e-mail and who it is from, it should be able to communicate this information to you immediately.

- Are always on

By default a wearable is always on and working, sensing, and acting. This is opposed to the normal use of pen-based PDAs, which normally sit in one's pocket and are only switched on when a task needs to be done.

This list, and indeed any general discussion of wearable computers, should be interpreted as guidelines rather than absolute. In particular, good wearable computing design depends greatly on the particular applications intended for the device.

3. Potential applications

With computer chips getting smaller and cheaper, the day will soon come when the desk, lap, and palm-top computer will all disappear into a vest pocket, wallet, shoe, or anywhere a spare centimeter or two is available. So, what kinds of applications can we expect to see when the bulk of the portable PC disappears into our clothing? As the processing power increases and the machines get smaller, the applications will be limited only by the quality of the sensory, input and output, and networking capability available. One thing is sure, there is likely to be a whole range of novel applications — consider the following.

- Proactive monitoring

With sensory devices that can monitor vital body signs, we can use wearables as the start of a powerful personal health monitor. The wearable can collate information on a regular basis, pass it on for processing and ensure that any symptoms that can lead to some medical condition can be tracked or identified at an early stage.

- Augmented reality [1]

With sensory devices that track the user's gaze or position, we are well placed to have applications that use the real world as a part of their interface. For example, an application might overlay objects in the real world with annotations describing the object [2], or draw the user's attention to certain items and guide them in particular directions using graphical overlays or spatially localised audio.

- Augmented intelligence

There are a range of data capture devices that can gather information about the user's environment and activities at any time — images, sound, temperature, light, location. These can be combined to form higher level sensors that might capture whether a user is inside/outside, on their own, talking to someone, etc. This contextual information can be utilised by a host of applications such as memory aids, guidance systems, proactive information gatherers, etc.

- Wearable workgroups and remote intelligence

With improvements in radio networks and bandwidth limitations reduced, we can expect groups to interact naturally wherever their physical location. Applications exist now where one can act as a remote on-the-spot agent, relaying information back to whoever needs that knowledge, for example on a building site [3], or remote evaluation of car insurance claims. The information need not be simply audio and video, it can extend to whatever can be sensed and forwarded. Thus, individuals could act as real-time sensors for surveys.

Until recently, computers have only had access to a user's current context within a computational task, but not outside of that environment. For example, a word-processor has access to the words currently typed, and perhaps files previously viewed. However, it has no way of knowing where its user is, whether she is alone or with someone, whether she is thinking or talking or reading, etc. Wearable computers give the opportunity to bring new sensors and technology into everyday life, such that these pieces of physical context information can be used by the wearable computer to provide proactive, context-dependent support to users.

4. Design needs for wearables

Making a portable computer or PDA and re-engineering it as a wearable computer is often not appropriate. Many of the design requirements for portables no longer apply in the wearable-computing environment. This section analyses a whole range of requirements covering input, output/display, power, and comfort.

4.1 User input devices

Traditional keyboards as input devices are not appropriate on the move — they rely on a steady surface and cannot be effectively used while walking. Traditional keyboards are also too large to be hidden from view or to be kept unobtrusive, which is important in many social situations.

One keyboard replacement currently in use is the Twiddler¹. This is a one-handed chord keyboard and mouse that once the chords are learnt, allows input at a rate of 50+ words per minute. However, it has to be attached to the user's hand, so if hands-free operation is needed for the particular application it is not appropriate. In many cases though this is an acceptable solution as it can be used on the move, is not particularly intrusive and quite robust. There are other keyboards that may be appropriate for wearables, such as the half-QWERTY [4] one-handed keyboard that exploits the symmetry of left and right hands in typing, and the BAT chording keyboard². These can be belt-mounted so users do not need to have it in their hand at all times.

Speech recognition systems [5] such as single word, or short phrase recognition systems are mature enough to be accurate for command-driven sequences or dictation in a controlled environment³. So long as it is not a problem if the wearable is controlled by speech, i.e. the user does not have to be quiet or have to control the system when in normal dialogue, then speech input is possible and allows completely hands-free operation.

More specialised input devices may be developed for particular applications, for example, from robust menu selection devices such as the input dial [6], through to touch pads, data gloves, gesture recognition systems [7], or radio controlled mouse.

4.2 Output devices

Visual

To look at a screen on the move, the displays have to be attached close and firmly to the eye. The solutions of current practical systems, although functional, leave much to be desired in terms of aesthetics. An example of a small head-mounted display is the Private eye display⁴ (see Fig 2 in section 5), which gives 720 × 280 monochrome resolution for very low power consumption. The majority of LCD head-up displays will give a crisp image from quarter-VGA to VGA quality, with either colour or 128-bit greyscale. Virtually all the approaches to head-mounted displays leave it clear that users have a display in front of them. However, there have been some recent advances in

embedding the display into glasses by Microoptical Inc⁵ as shown in Fig 1. This should to a large extent make head-mounted displays socially acceptable. Depending on the application, it may be entirely feasible to use displays that are not head-mounted, say wrist-mounted or on some pocket display such as the range of PDAs. In this case, there is much more flexibility in the positioning of the display and the ability to make it more discrete.

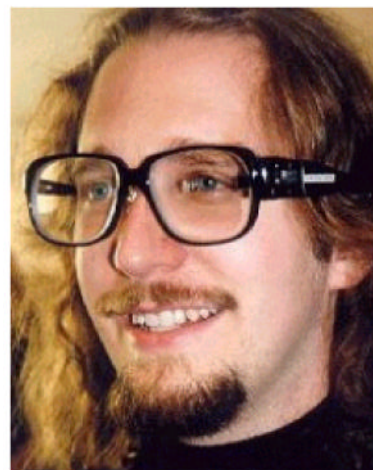


Fig 1 Thad Starner wearing the microoptical head-up display. A 320 × 240 pixel image is projected into the left eye.

Audio

Because audio does not detract the user in the same way as a screen or display interface, audio output is especially useful where the user is driving, involved in delicate operations, or may be visually impaired⁶. Roy et al [8] give a thorough overview of audio as both an input and output medium. Audio is also particularly useful for conveying peripheral information.

Touch

Tactile 'displays' may play an important role in wearable computers. We are all familiar with pagers or mobile phones that can be made to vibrate to bring the user's attention to new messages. This technique could be used as a simple direction 'display', with the appropriate device vibrating to point users in the right direction. More sophisticated tactile displays could be used to 'draw' images on the skin. Tan and Pentland [9] review these and other tactile displays.

4.3 Environment sensors

One key benefit of wearable computers will be their ability to make use of the immediate environment. Applications based on augmented reality and intelligence augmentation have already been briefly mentioned, but they

¹ www.handykey.com

² www.callamer.com/infogrip/bat.html

³ For example: IBM's ViaVoice, Dragon Systems' Naturally Speaking and Kurzweil's Voice Commands.

⁴ www.pedinc.com/PersonalDisplayDevice.htm

⁵ www.microoptical.corp.com

⁶ www.bham.ac.uk/alumni/magazine/pg27_96.htm

need some way of sensing the environment and the user's position within that environment to be effective. Crude position location can be achieved outside with GPS systems that can give position information to a few metres (with differential signals) together with speed and direction. As soon as the user moves inside a building the problem becomes more difficult — buildings have to be adorned with some kind of location beacons that can be picked up by the wearable [10]. Of course, the wearable user could manually update their position. Position is only one factor of potential sensors on a wearable. Simple sensors that measure temperature, humidity, noise levels, light levels, movement, etc, are also available. Combined with image and voice recognition systems these sensors can provide an excellent basis to model the user's environment in order to provide cues for context-based applications.

4.4 Networking

There are a number of existing technologies for general purpose network connections. For general roaming, cellular modems are popular in the US and give reasonable data rates (up to 19 kbit/s). In the UK and Europe, GSM is used to give 9.6 kbit/s, but shortly the GPRS (General Packet Radio Standard) will be available to give between 70 and potentially up to 170 kbit/s. For indoor use wireless LAN or DECT is appropriate. The Universal Mobile Telecommunications System (UMTS) may be the long-term approach, giving flexible bandwidth through its use of pico, micro and macro cell architecture.

The need for a network connection will vary depending on particular applications, but networking is essential for compute-hungry or information-hungry applications that cannot be achieved with processing and disk space local to the wearable.

4.5 Power

Power consumption provides a significant limitation to the application of wearable computing. More disk and processing power consumes more power, as does network connectivity and, to a lesser extent, the type of sensors used on the wearable. Where there is wireless network connectivity, local storage and processing power can be traded off for remote storage/processing. However, with relatively meagre requirements (processing power, hard disc, manual input devices and simple display) we can have wearable systems that use readily available high-quality lithium ion batteries that are relatively light and last for a good number of hours. For example, a Pentium P90 with 16 MB memory, 2.1 GB disk would consume about 14 W. We can expect much better performance with dedicated hardware, for example the 'Itsy' ⁷ project specifies a pocket computer based on the ARM risc

processor. Table 1 shows comparative power consumption for these processor architectures ⁸.

Table 1 Comparative processor power consumption.

Processor	Power (watts)
Pentium (100—200 MHz)	10—15
Pentium (w/VRT 100—166 MHz)	7
486/100	4—5
486/40	2—3
386	1—2
StrongArm 110 (160 MHz)	0.3
StrongArm 100 (200 MHz)	0.9

The wearable must be thought of in terms of the application first, then the appropriate interface technology built around that. All the choices of display, input device, networking, processing and power requirements become clearer when discussed in terms of particular applications. To substantiate this view, the next section looks at one application — the wearable remembrance agent.

5. The wearable remembrance agent

Current computer-based memory aids have been developed to make life easier for the computer, not for the person using them. For example, the two most common methods for accessing computer data is through filenames (forcing the user to recall the name of the file), and browsing (forcing the user to scan a list and recognise the name of the file). Both these methods are easy to program but require the user to do the brunt of the memory task themselves. Hierarchical directories or structured data such as calendar programs help only if the data itself is very structured, and break down whenever a file or a query does not fit into the redesigned structure. Similarly, key-word searches only work if the user can think of a set of words that uniquely identifies what is being searched for.

The remembrance agent (RA) [11], which has also been re-engineered in BT to link with Microsoft Word ⁹, watches what the user is reading or writing in a text editor and actively suggests personal e-mails, papers, or other text documents that may be relevant. This allows the user to focus on his primary task (in this case writing a report) and delegates the search for any supporting material to the remembrance agent. The system has been in daily use for over a year now, and the suggestions it produces are often quite useful. For example at BT, journal abstracts from the past several years have been indexed, and the RA used to suggest follow-up papers for users editing a document.

⁷ www.research.digital.com/wrl/itsy/

⁸ Taken from a posting to wear-hard@haven.org by R Paul McCarty (rp-mc@troi.cc.rochester.edu)

⁹ www.labs.bt.com/projects/agents/research/personal4.htm

The remembrance agent was ported to a wearable computer over a year ago. The wearable is based on the 'Lizzy'¹⁰ design, which in this case consisted of a 486 processor running Linux with a 'hat-top' display and twiddler as an input device; Fig 2 shows how the display is mounted, while Fig 3 shows a sample of the results from the remembrance agent. When the system was finally ported to a wearable computer even without modification, new applications became apparent. For example, when taking notes at a conference the remembrance agent will often suggest a document that leads to questions for the speaker. Because the wearable is taken everywhere, the RA can also offer suggestions based on notes taken during coffee breaks, where laptop computers cannot normally be used. Another advantage is that because the display is proactive, the wearer does not need to expect a suggestion in order to receive it. One common practice among the wearable users at conferences is to type in the name of every person met while shaking hands. The RA will occasionally remind the wearer that the person whose name was entered has actually been met before, and can even suggest the notes taken from that previous conversation. Of course, even more preferable is a wearable that knows who is in the area without the user having to type anything, through sensors such as automatic face recognition [12], or active badge systems.



Fig 2 The 'hat-top' display for the remembrance agent.

Human memory does not operate in a space of query-response pairs. On the contrary, the context of a remembered episode provides lots of cues for recall later. These cues include the physical location of an event, who

¹⁰ wearables. www.media.mit.edu/projects/wearables/lizzy

was there, what was happening at the same time, and what happened immediately before and after [13]. This information both helps us recall events given a partial context, and to associate our current environment with past experiences that might be related. The wearable remembrance agent has been augmented to use, when available, automatically detected physical context to help determine relevant information. This context information is used both to tag information, and later for retrieval (see also Salton [14]). Notes taken on the wearable are therefore tagged with context information and stored for later retrieval. In suggestion-mode, the wearer's current physical context is used to find relevant information. If sensor data is not available (for example if no active-badge system is in use) the wearer can still type in additional context information themselves. The current version of the wearable remembrance agent uses five context cues to produce relevant suggestions:

- wearer's physical location — this information could be provided by GPS, an indoor location system, or a location entered explicitly by the user on the chording keyboard,
- people who are currently around — this information can come from an active badge system, another person's wearable computer, or again can be entered by the wearer,
- subject field, which can be entered by the wearer as an extra tag, or in indexing can be extracted from header fields such as the subject line in e-mail,
- date and time-stamp — these can be stamped on to note files with a single chord on the keyboard, or can be extracted from more structured data (in retrieval, this information comes from the system clock),
- the information itself (the body of the note) — this is turned into a word-vector for later keyword analysis (in retrieval, this information comes from whatever the wearer is currently reading or writing on the heads-up display).

6. Usability

Sometimes a suggestion summary line can be enough to jog a memory, with no further look up necessary (see Fig 3). However, it is often desirable to look up the complete reference being summarised. In these cases, a single chord can be hit on the chording keyboard to bring up

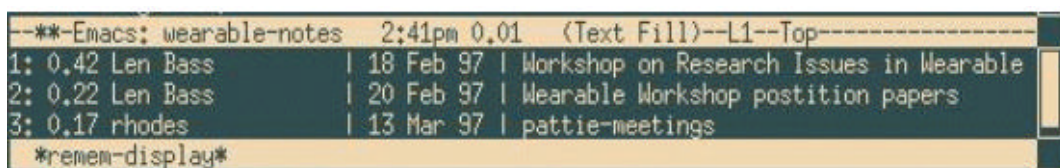


Fig 3 Example output from the remembrance agent. Each line shows a suggestion. First the degree of match is shown followed by the author, date and subject of the suggestion.

any of the suggested references in the main buffer. If the suggested file is large, the RA will automatically jump to the most relevant point in the file before displaying it. The system has been in daily use for over six months, and several design issues are already apparent from using this prototype. These issues will help drive the next set of revisions.

The biggest design trade-off with the RA is between making continuous suggestions versus only occasionally flashing suggestions in a more obtrusive way. The continuous display was designed to be as tolerant of false positives as possible, and to distract the wearer from the real world as little as possible. The continuous display also allows the wearer to receive a new suggestion literally in the blink of an eye rather than having to fumble with a keyboard or button. However, because suggestions are displayed even when no especially relevant suggestions are available, the wearer has a tendency to distrust the display. After a few weeks of use our experience suggests that the wearer tends to ignore the display except when they are looking at the screen anyway, or when they already realise that a suggestion might be available. The next version of the RA will cull low-relevancy hits entirely from the display, leaving a variable-length display with more trustworthy suggestions.

Furthermore, a 'visual bell' that flashes on the screen several times will accompany notifications that are judged to be too important to miss (for example, notification that a scheduled event is about to happen). This flashing is already being used in a wearable communications system on the current heads-up mounted display, and has been satisfactory in getting the wearer's attention in most cases. Another lesson learned from the interface for this communications system is that the screen should radically change when an important message is available. This way the wearer need not read any text to see if there is an important alert. Currently, the communications system prints a large reverse-video line across the lower half of the screen, which is used to quickly determine if a message has arrived.

Another trade-off has been made between showing lots of text on the screen versus showing only the most important text in larger fonts. The current design shows an entire 80 column by 25-row screen, but this often produces too much text for a wearer to scan while still trying to carry on a conversation. Future versions will experiment with variable font size and animated typography [15].

Habituation has been an issue with the current system. Currently the user can set by hand how the RA should bias different features of physical context, but in the future biases should be automatically changed according to the user's context. For example, when a user first enters a room, their new location should be an important factor in choosing information to show. After a few minutes, if that

location has not changed, the RA should bias towards newer information.

7. Related work

Probably the closest system to the wearable RA is the Forget-me-not system developed at The Rank Xerox Research Centre [16]. The Forget-me-not is a PDA system that records where its user is, who they are with, who they phone, and other such autobiographical information, and stores it in a database for later query. It differs from the RA in that the RA looks at and retrieves specific textual information (rather than just a diary of events), and the RA has the capacity to be proactive in its suggestions as well as answer queries.

Several systems also exist to provide contextual cues for managing information on a traditional desktop system. For example, the Lifestreams project provides a complete file management system based on time-stamp [17]. It also provides the ability to tag future events, such as meeting times that trigger alarms shortly before they occur.

Finally, several systems exist to recommend Web pages based on the pages a user is currently browsing [18, 19].

8. Conclusions

Building an effective wearable computing system needs careful consideration of the display technology, and the input and sensing devices. With the wearable RA, we were in some ways lucky in that the Lizzy was available and had users who were happy to put up with less than ideal I/O, which made it easier for us to design the RA and allowed us to concentrate more on the application side and how that could be improved.

The wearable RA has certainly been useful to date, with the most helpful feature being its ability to search a body of messages for appropriate text, and to search on the person speaking.

It is not difficult to see that in more specialised, mobile communities, this context-based reminding will prove invaluable, and that the motivating scenario described in the introduction is technically feasible.

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Barry Crabtree joined BT in 1980 after completing a Physics degree at Bath University. After a number of years in software development he moved into the field of artificial intelligence (AI) and helped begin BT's systems in automated diagnosis and workforce management. He moved on to work in distributed AI, optimisation problems and adaptive systems. He is currently a technical advisor in Applications, Research and Technology (ART), and has particular interest in intelligent agents and wearable computers.



Bradley J Rhodes works in the software agents group at the MIT Media Lab, and his particular focus is on designing 'remembrance agents' that provide just-in-time information based on their user's current environment. The work tends to cross several disciplines, from artificial intelligence and cognitive science to human/computer interaction and design. Before his current research he worked on building autonomous characters for interactive fiction and automatic storytelling systems.